INVERTIS UNIVERSITY BAREILLY

Established by Govt. of U.P. u/s 2F of UGC Act, 1956 vide U.P. Act 22 of 2010.

BCS-055	DISCRETE STRUCTURES	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To develop logical thinking and its application to computer science (to emphasize the importance of proving statements correctly.
CO2	To Have substantial experience to comprehend formal logical arguments.
CO3	To express mathematical properties formally via the formal language of propositional logic and predicate logic.
CO4	To understand basic mathematical objects such as sets, functions, and relations and will also be able to verify simple mathematical properties that these objects possess.
CO5	The subject enhances one's ability to reason and ability to present a coherent and mathematically accurate argument.

DETAILED SYLLABUS

MODULE-I

Set Theory: Introduction, Combination of sets, Multisets, Ordered pairs. Proofs of some general identities on sets. **Relations:** Definition, Operations on relations, Properties of relations, Composite Relations, Equality of relations, Recursive definition of relation, Order of relations.
Functions: Definition, Classification of functions, Operations on functions, Recursively defined functions. Natural Numbers: Introduction, Mathematical Induction, Proof Methods, Proof by contradiction. **Algebraic Structures:** Definition, Groups, Subgroups and order, Cyclic Groups, Lagrange's theorem, Normal Subgroups, Permutation and Symmetric groups, Group Homeomorphisms, Definition and elementary properties of Rings and Fields.

MODULE-II

Partial order sets: Definition, Partial order sets, Combination of partial order sets, Hasse diagram. **Lattices:** Definition, Properties of lattices – Bounded, Complemented, Modular and Complete lattice. **Boolean algebra:** Introduction, Axioms and Theorems of Boolean algebra, Algebraic manipulation of Boolean expressions. Simplification of Boolean Functions, Karnaugh maps, Logic gates, Digitalcircuits and Boolean algebra.
Propositional Logic: Proposition, well formed formula, Truth tables, Tautology, Satisfiability, Contradiction, Algebra of proposition, Theory of Inference.
Predicate Logic: First order predicate, well formed formula of predicate, quantifiers, Inference theory of predicate logic.

MODULE-III

Trees : Definition, Binary tree, Binary tree traversal, Binary search tree.
Graphs: Definition and terminology, Representation of graphs, Multigraphs, Bipartite graphs, Planar graphs, Isomorphism and Homeomorphism of graphs, Euler and Hamiltonian paths, Graph coloring, Recurrence Relation, Method of solving recurrences.

Text Book*:*

1. Koshy, Discrete Structures, Elsevier Pub. 2008

Reference Books:

1. Kenneth H. Rosen, Discrete Mathematics and Its Applications, 6/e, McGraw-Hill, 2006.
2. B. Kolman, R.C. Busby, and S.C. Ross, Discrete Mathematical Structures, 5/e, Prentice Hall, 2004.
3. E.R. Scheinerman, Mathematics: A Discrete Introduction, Brooks/Cole, 2000.
4. R.P. Grimaldi, Discrete and Combinatorial Mathematics, 5/e, Addison Wesley, 2004.
5. J.P. Tremblay& R. Manohar, "Discrete Mathematical Structure with Applications to Computer Science" Mc.Graw Hill, 1975

Course Outcomes: After the completion of the course the student will be able to:

CO1	Be able to construct simple mathematical proofs and possess the ability to verify them.
CO2	Have substantial experience to comprehend formal logical arguments.
CO3	Be skillful in expressing mathematical properties formally via the formal language of propositional logic and predicate logic.
CO4	Be able to specify and manipulate basic mathematical objects such as sets, functions, and relations and will also be able to verify simple mathematical properties that these objects possess.
CO5	Gain experience in using various techniques of mathematical induction (weak, strong and structural induction) to prove simple mathematical properties of a variety of discrete structures.